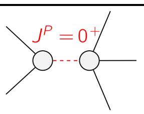
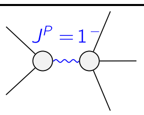
$$
\begin{array}{c|cc|cc|c|c}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#2}_{0^+} & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{3^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}\dagger & \Upsilon_1 k^2 + \overset{(2)}{\kappa}_2 & \Upsilon_1 k^2 + \overset{(2)}{\kappa}_2 & & & & \\
\mathcal{K}^{\#2}_{0^+}\dagger & \Upsilon_1 k^2 + \overset{(2)}{\kappa}_2 & \Upsilon_1 k^2 + \overset{(2)}{\kappa}_2 & & & & \\
\mathcal{K}^{\#1}_{1^-}\dagger^{\alpha} & & & \frac{1}{3}\,(k^2\,\overset{(4)}{\kappa}_2 + \overset{(2)}{\kappa}_2) & \frac{1}{3}\,(\sqrt{5}\,k^2\,\overset{(4)}{\kappa}_2 + \sqrt{5}\,\overset{(2)}{\kappa}_2) & & \\
\mathcal{K}^{\#2}_{1^-}\dagger^{\alpha} & & & \frac{1}{3}\,(\sqrt{5}\,k^2\,\overset{(4)}{\kappa}_2 + \sqrt{5}\,\overset{(2)}{\kappa}_2) & \frac{1}{3}\,(5\,k^2\,\overset{(4)}{\kappa}_2 + 5\,\overset{(2)}{\kappa}_2) & & \\
\mathcal{K}^{\#1}_{2^+}\dagger^{\alpha\beta} & & & & & 0 & \\
\mathcal{K}^{\#1}_{3^-}\dagger^{\alpha\beta\chi} & & & & & & 0
\end{array}
$$

$$
\begin{array}{c|cc|cc|c|c}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#2}_{0^+} & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{3^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}\dagger & \frac{1}{2\,\mathrm{Det}\,(0^+)} & \frac{1}{2\,\mathrm{Det}\,(0^+)} & & & & \\
\mathcal{J}^{\#2}_{0^+}\dagger & \frac{1}{2\,\mathrm{Det}\,(0^+)} & \frac{1}{2\,\mathrm{Det}\,(0^+)} & & & & \\
\mathcal{J}^{\#1}_{1^-}\dagger^{\alpha} & & & \frac{1}{12\,(k^2\,\overset{(4)}{\kappa}_2 + \overset{(2)}{\kappa}_2)} & \frac{\sqrt{5}}{12\,(k^2\,\overset{(4)}{\kappa}_2 + \overset{(2)}{\kappa}_2)} & & \\
\mathcal{J}^{\#2}_{1^-}\dagger^{\alpha} & & & \frac{\sqrt{5}}{12\,(k^2\,\overset{(4)}{\kappa}_2 + \overset{(2)}{\kappa}_2)} & \frac{5}{12\,(k^2\,\overset{(4)}{\kappa}_2 + \overset{(2)}{\kappa}_2)} & & \\
\mathcal{J}^{\#1}_{2^+}\dagger^{\alpha\beta} & & & & & 0 & \\
\mathcal{J}^{\#1}_{3^-}\dagger^{\alpha\beta\chi} & & & & & & 0
\end{array}
$$

Abbreviations used in matrices
$\Upsilon_1 == \overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2 \;\&\&\; \mathrm{Det}\,(0^+) == 2\,(k^2\,(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2) + \overset{(2)}{\kappa}_2)$
Lagrangian
$\overset{(2)}{\kappa}_2\;\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\;\mathcal{K}_{\beta}{}^{\chi}{}_{\chi} + \overset{(4)}{\kappa}_1\,\partial_{\beta}\mathcal{K}_{\chi}{}^{\delta}{}_{\delta}\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_2\,\partial_{\chi}\mathcal{K}_{\beta}{}^{\delta}{}_{\delta}\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#2}_{0^+} == 3\,\mathcal{J}^{\#1}_{0^+}$	1	$2\,\partial_{\chi}\partial_{\beta}\partial_{\alpha}\mathcal{J}^{\alpha\beta\chi} == \partial_{\chi}\partial^{\chi}\partial_{\beta}\mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta}$
$\mathcal{J}^{\#2\,\alpha}_{1^-} == 5\,\mathcal{J}^{\#1\,\alpha}_{1^-}$	3	$6\,\partial_{\delta}\partial_{\chi}\partial_{\beta}\partial^{\alpha}\mathcal{J}^{\beta\chi\delta} + \partial_{\delta}\partial^{\delta}\partial_{\chi}\partial^{\chi}\mathcal{J}^{\alpha\beta}{}_{\beta} == \partial_{\delta}\partial^{\delta}\partial_{\chi}\partial^{\alpha}\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + 6\,\partial_{\delta}\partial^{\delta}\partial_{\chi}\partial_{\beta}\mathcal{J}^{\alpha\beta\chi}$
$\mathcal{J}^{\#1\,\alpha\beta}_{2^+} == 0$	5	$k^{\chi}\;(\partial_{\epsilon}\partial_{\chi}\partial^{\beta}\partial^{\alpha}\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 2\,\partial_{\epsilon}\partial_{\delta}\partial^{\beta}\partial^{\alpha}\mathcal{J}_{\chi}{}^{\delta\epsilon} - 3\,\partial_{\epsilon}\partial_{\delta}\partial_{\chi}\partial^{\alpha}\mathcal{J}^{\beta\delta\epsilon} -$ $3\,\partial_{\epsilon}\partial_{\delta}\partial_{\chi}\partial^{\beta}\mathcal{J}^{\alpha\delta\epsilon} + 3\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial_{\chi}\mathcal{J}^{\alpha\beta\delta} + \eta^{\alpha\beta}\;\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\partial_{\chi}\mathcal{J}^{\delta\epsilon\phi} - \eta^{\alpha\beta}\;\partial_{\phi}\partial^{\phi}\partial_{\epsilon}\partial_{\chi}\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}) == 0$
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{3^-} == 0$	7	$k^{\delta}\;k^{\epsilon}\;(3\,\partial_{\epsilon}\partial^{\chi}\partial^{\beta}\partial^{\alpha}\mathcal{J}_{\delta}{}^{\phi}{}_{\phi} - \partial_{\epsilon}\partial_{\delta}\partial^{\beta}\partial^{\alpha}\mathcal{J}^{\chi\phi}{}_{\phi} - \partial_{\epsilon}\partial_{\delta}\partial^{\chi}\partial^{\alpha}\mathcal{J}^{\beta\phi}{}_{\phi} - \partial_{\epsilon}\partial_{\delta}\partial^{\chi}\partial^{\beta}\mathcal{J}^{\alpha\phi}{}_{\phi} + 2\,\partial_{\phi}\partial^{\chi}\partial^{\beta}\partial^{\alpha}\mathcal{J}_{\delta\epsilon}{}^{\phi} -$ $4\,\partial_{\phi}\partial_{\epsilon}\partial^{\beta}\partial^{\alpha}\mathcal{J}^{\chi}{}_{\delta}{}^{\phi} - 4\,\partial_{\phi}\partial_{\epsilon}\partial^{\chi}\partial^{\alpha}\mathcal{J}^{\beta}{}_{\delta}{}^{\phi} - 4\,\partial_{\phi}\partial_{\epsilon}\partial^{\chi}\partial^{\beta}\mathcal{J}^{\alpha}{}_{\delta}{}^{\phi} + 5\,\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\beta\chi\phi} + 5\,\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\partial^{\beta}\mathcal{J}^{\alpha\chi\phi} +$ $5\,\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\partial^{\chi}\mathcal{J}^{\alpha\beta\phi} - 5\,\partial_{\phi}\partial^{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\alpha\beta\chi} - \eta^{\beta\chi}\;\partial_{\gamma}\partial_{\epsilon}\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\phi}{}_{\phi}{}^{\gamma} - \eta^{\alpha\chi}\;\partial_{\gamma}\partial_{\epsilon}\partial_{\delta}\partial^{\beta}\mathcal{J}^{\phi}{}_{\phi}{}^{\gamma} - \eta^{\alpha\beta}\;\partial_{\gamma}\partial_{\epsilon}\partial_{\delta}\partial^{\chi}\mathcal{J}^{\phi}{}_{\phi}{}^{\gamma} +$ $\eta^{\beta\chi}\;\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial^{\alpha}\mathcal{J}_{\delta}{}^{\phi\gamma} + \eta^{\alpha\chi}\;\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial^{\beta}\mathcal{J}_{\delta}{}^{\phi\gamma} + \eta^{\alpha\beta}\;\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial^{\chi}\mathcal{J}_{\delta}{}^{\phi\gamma} - \eta^{\beta\chi}\;\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\alpha\phi\gamma} - \eta^{\alpha\chi}\;\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\beta\phi\gamma} -$ $\eta^{\alpha\beta}\;\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\chi\phi\gamma} + \eta^{\beta\chi}\;\partial_{\gamma}\partial^{\gamma}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\alpha\phi}{}_{\phi} + \eta^{\alpha\chi}\;\partial_{\gamma}\partial^{\gamma}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\beta\phi}{}_{\phi} + \eta^{\alpha\beta}\;\partial_{\gamma}\partial^{\gamma}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\chi\phi}{}_{\phi}) == 0$
Total # constraint(s):	16	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$-\frac{\overset{(2)}{\kappa}_2}{\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2}$	$\frac{1}{2\,(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2)}$
	3	$-\frac{\overset{(2)}{\kappa}_2}{\overset{(4)}{\kappa}_2}$	$-\frac{1}{2\,\overset{(4)}{\kappa}_2}$
Resolved unitarity condition(s):		(Demonstrably impossible)	